

ANALYSTLAB AFRICA
Experience Lab Internship Programme



HealthConnect Clinic
Assumptions, Limitations, Risks & Dependencies Report

By
Ogheneughwe Godwin AKISE

Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance
and Healthcare Support Using Data and AI
Track: Data Science

Prepared for the AnalystLab Africa Experience Lab
Week 7 Submission – Consolidated Review of Weeks 4–6

Table of Contents

1. Issues that have been resolved
 2. Issues that remain unresolved
 3. New issues discovered
 4. New cross-track dependencies
 5. Technical limitations
 6. Data limitations
 7. Modelling limitations
 8. AI safety/reliability concerns
 9. Integration challenges
 10. Potential impact on the project
 11. Recommended mitigation
- References

This report is a Week 7 Data Science track deliverable for the HealthConnect Clinic Experience Lab engagement. It consolidates the assumptions, limitations, risks and dependencies identified across the track's Week 4 Risk and Dependency Register, the Week 5 Review of Week 4 Foundation, the Week 5 Project Execution & Progress Package, and the Week 6 Model Improvement & Validation Report, presenting a single, current picture of what has been resolved, what remains open, and what has newly emerged, ahead of Week 7 testing and the Week 8 Final Presentation.

Maintaining a consolidated, actively reviewed register of this kind is standard project management practice, consistent with the Project Management Institute's guidance on identifying, analysing and responding to project risk throughout a project's life cycle (Project Management Institute [PMI], 2021), and with the NIST AI Risk Management Framework's guidance that AI-specific risks — including bias, reliability and safety — should be tracked and managed continuously rather than assessed once (Tabassi, 2023). This report should be read alongside, rather than instead of, the four source documents named above; it does not repeat their full technical content, only the items relevant to assumptions, limitations, risks and dependencies.

Key Assumptions Carried Forward

The following assumptions, first stated in Week 4, remain in effect and continue to shape how findings from this project should be interpreted:

- The HealthConnect Appointment Dataset is fictional and synthetic; all findings are illustrative of methodology rather than clinically validated, and this limitation is restated explicitly in every report produced by this track.
- The model is intended to support clinic staff in prioritising outreach for high-risk appointments, not to automate scheduling or intervention decisions.
- Only information genuinely available at the time of booking should be used as a model input; this assumption underpins the exclusion of `waiting_time_minutes` and the leakage-safe design of the Week 6 threshold-optimisation work.
- HealthConnect has not yet supplied a formal cost weighting between a missed no-show (False Negative) and an unnecessary patient contact (False Positive); until it does, this track assumes that reducing missed appointments should not come at a disproportionate cost to the patient support experience, consistent with the Week 4 problem framing.
- Patient identifiers, while not present in the current processed feature matrices, are assumed to be recoverable from the original dataset if required for patient-grouped validation in Week 7 (see Sections 3 and 4).

1. Issues that have been resolved

The following issues, first identified in Weeks 4 and 5, have been substantively resolved by the end of Week 6:

Issue	Originally Identified	Resolution
Repeated-patient leakage risk — 5,000 appointments across only 1,696 patients risked patient overlap between train and test sets under a standard random split.	Week 5	Resolved: a patient-aware <code>GroupShuffleSplit</code> (grouped by <code>patient_id</code>) was adopted, with zero patient overlap between the

Issue	Originally Identified	Resolution
		training and test sets confirmed and carried forward unchanged into Week 6.
waiting_time_minutes leakage ambiguity (R1) — unclear whether the field was available at booking time or only after the appointment.	Week 4	Mitigated: excluded from the feature set as a precaution; a Week 5 t-test found no significant group difference ($p = 0.781$), supporting the decision. Formal data-owner confirmation is still outstanding (Section 2).
Missing threshold-sensitivity evidence — an earlier version of the Week 6 report flagged that narrative references to threshold_df / threshold_df_refined tables could not be verified against an executed notebook cell.	Week 6	Resolved: the revised Week 6 notebook constructed leakage-safe threshold_df_refined and threshold_business_df tables from out-of-fold cross-validated probabilities, and this evidence is now fully reported in the Week 6 Model Improvement & Validation Report, Section 6.3.
Feature-refinement hypotheses left unvalidated — three interaction features engineered from the error analysis had not yet been confirmed to improve performance without overfitting.	Week 6	Resolved: the Refined Logistic Regression was retrained on the refined feature set and compared fairly against the unmodified baseline, confirming a modest but genuine ROC-AUC/PR-AUC gain with no material increase in overfitting (Week 6 report, Sections 5 and 5.1).
Ambiguity between the Week 4 proposed chronological split and an alternative patient-aware split.	Week 4 → 5	Resolved: the chronological split was deliberately superseded by a patient-aware grouped split, documented as an explicit, reasoned decision rather than left ambiguous (Week 5 Project Summary, Section 7).

2. Issues that remain unresolved

The following issues, first identified in Weeks 4 or 5, remain open as of the end of Week 6 and require action in Week 7:

Issue	Originally Identified	Current Status
Cancellation-exclusion definition (R2) — excluding Cancelled appointments from the target definition may not match the clinic's actual operational priorities.	Week 4	Open; not yet confirmed with HealthConnect stakeholders.

Issue	Originally Identified	Current Status
Subgroup fairness untested (R3) — model performance across demographic and access-related subgroups has not been formally assessed.	Week 4	Open; deferred from Week 5 to Week 6 to Week 7 (Week 6 report, Section 12.3). No fairness testing has yet been performed on the Refined Logistic Regression or its provisional 0.45 threshold.
Sunday-appointment anomaly unescalated (R7) — 737 appointments (14.7%) are scheduled on Sundays despite the Clinic Knowledge Base stating the clinic is closed that day.	Week 5	Open; mitigated with an engineered flag, but escalation to the data owner or mentor (planned as Week 6 action T19) has not been confirmed as completed.
README/notebook split-method description (R8, T14) — the README's description of the train/test split method has not been confirmed as updated to reflect the patient-aware method actually used.	Week 5	Open; the substantive metrics-reporting inconsistency was resolved in Week 5, but this specific documentation item was not confirmed as corrected in Week 6.
waiting_time_minutes formal confirmation (R1, D2) — the exclusion decision is well supported internally but has not been formally confirmed by the data owner.	Week 4	Open; internally mitigated, externally unconfirmed.
Cross-track syncs with Machine Learning Engineering, Project Management and Generative AI (R6) — proposed in the Week 5 Track Coordination Plan.	Week 4	Partially open; only the Data Analytics exchange has a documented, auditable evidence artefact (Week 6 report, Section 11.4). The other three syncs have not been confirmed as completed.
Provisional operating threshold not yet stakeholder-confirmed — the HealthConnect-aligned 0.45 threshold was selected using training-only evidence, not a HealthConnect-supplied cost weighting.	Week 6	Open; explicitly flagged as provisional pending Week 7 stakeholder validation (Week 6 report, Sections 7 and 12.1).

3. New issues discovered

The following issues were newly discovered during Week 6 and had not been documented in the Week 4 or Week 5 registers:

ID	Issue	Description
NI1	Absence of patient identifiers in processed feature matrices	The processed Week 5/6 feature matrices (X_train_refined, X_test_refined) do not retain patient_id, so the Week 6 threshold-selection cross-validation could only use ordinary stratified k-fold rather than the patient-grouped separation used for the original train/test split —

ID	Issue	Description
		reintroducing a milder form of the patient-leakage risk that Week 5 had otherwise resolved for model training.
NI2	Untuned tree-based models overfit substantially	Random Forest (train ROC-AUC 0.9903 vs. test 0.6637) and Histogram Gradient Boosting (train ROC-AUC 0.8766 vs. test 0.6615) both showed severe train–test performance gaps with default hyperparameters, meaning neither can currently be treated as a reliable alternative to Logistic Regression.
NI3	Feature-engineering gain is small and not yet proven stable	The Refined Logistic Regression's improvement over the Week 5 baseline (+0.0017 ROC-AUC, +0.0037 PR-AUC) is small enough that it has not yet been demonstrated to be robust to the specific train/test split, rather than a sample-specific artefact.
NI4	No formal cost weighting supplied by HealthConnect	Threshold selection had to proceed using an internally defined intervention-efficiency rule (Extra False Positives per False Negative Avoided) rather than a HealthConnect-confirmed relative cost of a missed appointment versus an unnecessary patient contact — a dependency not previously documented as a risk.

4. New cross-track dependencies

Building on the Week 4 dependency register (D1–D5), the following cross-track dependencies were newly identified, or newly evidenced, during Week 6:

ID	Dependency	Track(s)	Status
D6	The Machine Learning Engineering track's Week 7 integration work now depends on the specific Week 6 artefacts saved for hand-off (refined_lr_model_wk6.pkl, the 38-feature refined schema, and the frozen provisional 0.45 threshold), not only the Week 4 target/feature definitions.	Machine Learning Engineering	Open — artefacts saved; formal hand-off not yet confirmed.
D7	Week 7 fairness and threshold-confirmation work depends on HealthConnect and/or the Project Management track supplying a formal false-positive tolerance or cost weighting.	Project Management / HealthConnect stakeholders	Open.
D8	Any future patient-grouped validation of the Week 6 threshold work depends on patient_id being recovered or re-attached to the processed feature matrices.	Data Science (internal)	Open.
D9	The Data Analytics cross-track exchange	Data Analytics	Satisfied —

ID	Dependency	Track(s)	Status
	now has a concrete, auditable evidence artefact (a shared dashboard file) rather than only a narrative description.		evidence recorded in the Week 6 report, Section 11.4.

5. Technical limitations

- Cross-validation used for Week 6 threshold selection could not preserve patient-level grouping (Section 3, NI1), a methodological gap inherited from the absence of patient_id in the saved processed matrices.
- No automated experiment-tracking or model-registry tooling is in place; model artefacts are currently version-managed manually (.pkl and .csv files saved per week), which increases the risk of the kind of documentation/code inconsistency already encountered once (R8).
- Random Forest and Histogram Gradient Boosting have not been hyperparameter-tuned or regularised, so their true technical potential relative to Logistic Regression remains unknown (Week 6 report, Section 5.1).
- Reproducibility safeguards introduced in Week 6 (explicit assertion checks confirming reloaded artefacts match the Week 5 baseline) are a genuine strength, but formal automated testing — for example, unit tests on the feature-engineering pipeline — is not yet in place.

6. Data limitations

- The HealthConnect Appointment Dataset is fictional and synthetic (R4); findings should be treated as illustrative of methodology rather than clinically validated, an assumption accepted since Week 4 and unchanged.
- Feature completeness: no socio-economic status, specific medical condition, or transportation-access data is available, which may constrain the model's predictive ceiling (Week 6 report, Section 10.1).
- The Sunday-appointment anomaly (R7) remains unresolved at the data-quality level; whether it reflects a data error or an undocumented clinic exception is still unconfirmed.
- waiting_time_minutes remains excluded from the feature set pending formal confirmation of its definition, representing a source of potentially useful signal that is currently, at minimum, deferred.
- The dataset's relatively small size (5,000 appointments; 1,696 unique patients) limits the granularity at which subgroup fairness testing (Section 8) can be performed with statistical confidence.

7. Modelling limitations

- Linearity: even with three engineered interaction terms, the Refined Logistic Regression may not capture more complex, non-linear relationships between features and no-show behaviour.

- Modest, not-yet-validated improvement: the ROC-AUC/PR-AUC gain over the Week 5 baseline is small and requires robust cross-validation to confirm it is not sample-specific (Section 3, NI3).
- Untuned non-linear alternatives: Random Forest and Histogram Gradient Boosting overfit substantially with default hyperparameters and cannot currently be considered as replacement candidates (Section 3, NI2).
- Threshold dependency: classification performance is highly sensitive to the chosen threshold; the provisional 0.45 threshold, while now evidence-based, has not been confirmed against a HealthConnect-defined cost structure (Section 3, NI4).
- Reliance on historical, potentially non-stationary data: changes in patient demographics, booking behaviour or clinic policy could cause drift, and no production monitoring plan has yet been implemented — only proposed (Week 6 report, Section 12.5).

8. AI safety/reliability concerns

The concerns below are framed with reference to the NIST AI Risk Management Framework's core functions of governing, mapping, measuring and managing AI risk throughout a system's life cycle (Tabassi, 2023), which this track adopted as a reference framework from Week 4 onward.

- Algorithmic bias and fairness: subgroup performance (e.g., by gender, age group, or distance to clinic as an access proxy) has not yet been formally tested; this remains open since Week 4 (R3) and is now more pressing because the provisional 0.45 threshold changes who is flagged for intervention.
- Automation bias and over-reliance: the model is intended to support, not replace, clinical and administrative judgement, but no human-in-the-loop review protocol has yet been formally documented for triggered interventions, beyond being proposed as a mitigation.
- Patient trust and perceived surveillance: the HealthConnect-aligned 0.45 threshold increases False Positives on the holdout test set (225 versus 176 at the default threshold), meaning more attending patients would be flagged and potentially contacted unnecessarily; this trade-off has not yet been validated with HealthConnect stakeholders or assessed for its effect on patient trust.
- Data privacy and security: patient data used for training and inference must continue to be handled in line with applicable privacy and healthcare-data obligations; no separate privacy or security review has been documented to date.
- Reliability under distribution shift: no drift-detection or monitoring mechanism is yet operational, so any degradation in reliability after deployment would not currently be automatically detected.

9. Integration challenges

- Cross-track synchronisation remains incomplete: only the Data Analytics exchange has a documented, auditable evidence artefact; the Machine Learning Engineering, Project Management and Generative AI syncs proposed in the Week 5 Track Coordination Plan have not been confirmed as completed.
- No formally agreed hand-off format with the Machine Learning Engineering track: the Week 6 artefacts (model file, refined feature CSVs, frozen threshold) were saved locally,

but no integration contract — for example, expected input schema or inference-latency requirements — has yet been documented jointly with that track.

- The Generative AI track's Clinic Knowledge Base states the clinic is closed on Sundays, which conflicts with the unresolved Sunday-appointment anomaly (R7); until this is escalated and resolved, the two tracks are effectively working from inconsistent assumptions about clinic operating hours.
- Documentation debt (the README's uncorrected split-method description) creates a risk that a track relying on the notebook's narrative documentation, rather than its executed code, could work from an inaccurate description of the modelling methodology.

10. Potential impact on the project

- If subgroup fairness is not tested before any pilot or deployment decision, HealthConnect risks a model that under-serves or over-targets specific patient groups, with reputational and ethical consequences for the wider Experience Lab project.
- If the provisional 0.45 threshold is adopted operationally without stakeholder confirmation, HealthConnect could face an intervention burden (225 unnecessary contacts observed on the holdout set) that exceeds actual outreach capacity, or could otherwise retain the default threshold's higher missed-no-show rate unnecessarily.
- Continued delay in cross-track synchronisation (Machine Learning Engineering, Project Management, Generative AI) increases the risk that downstream integration work is designed against outdated Week 4 assumptions rather than the current, refined Week 6 feature set and model artefacts.
- Unresolved data-quality questions (the Sunday-appointment anomaly and `waiting_time_minutes`) leave residual uncertainty in the underlying dataset that could affect confidence in any headline result presented at the Week 8 Final Presentation.
- Because the Week 6 performance gain over the baseline is small, there is a risk that Week 7 robustness testing could show the improvement is not stable, which would require re-evaluation of the candidate-model recommendation and the project's technical narrative ahead of final presentation.

11. Recommended mitigation

The following mitigations are recommended, prioritised by the severity of potential impact set out in Section 10:

Priority	Issue / Risk	Recommended Mitigation	Owner	Target
High	Subgroup fairness untested (R3)	Conduct formal fairness testing (e.g., demographic parity, equal opportunity, predictive parity) across gender, age group and distance-based access proxy before any deployment discussion.	Data Science	Week 7
High	Provisional threshold not confirmed (NI4)	Present the Week 6 <code>final_threshold_comparison</code> evidence to HealthConnect	Data Science / Project Management	Week 7

Priority	Issue / Risk	Recommended Mitigation	Owner	Target
		stakeholders and obtain a formal false-positive tolerance or cost weighting.		
High	Sunday-appointment anomaly unescalated (R7)	Formally escalate to the data owner or mentor for confirmation, coordinating with the Generative AI track given the shared Clinic Knowledge Base dependency.	Data Science / Project Management	Week 7 (overdue from Week 6)
Medium	Cross-track syncs incomplete (R6)	Schedule and document formal syncs with Machine Learning Engineering, Project Management and Generative AI, each with a recorded evidence artefact, as achieved with Data Analytics.	Data Science / Project Management	Week 7
Medium	Patient identifiers absent from processed matrices (NI1)	Recover or re-attach patient_id to the processed feature matrices (without using it as a model feature) specifically to enable patient-grouped cross-validation of the threshold-selection work.	Data Science	Week 7
Medium	Untuned tree-based models (NI2)	Perform systematic hyperparameter tuning and regularisation of Random Forest and Histogram Gradient Boosting before drawing conclusions about model-family suitability.	Data Science	Week 7/8 (if scope allows)
Medium	README/ documentation inconsistency (R8)	Correct the README's description of the train/test split method to match the patient-aware method actually used.	Data Science	Week 7
Low	waiting_time_minutes formal confirmation (R1)	Obtain written confirmation from the data owner of the field's definition, even though the feature is already excluded, to close the item formally.	Data Science	Week 7/8
Low	Cancellation-exclusion definition (R2)	Confirm with HealthConnect stakeholders whether excluding Cancelled appointments matches operational priorities.	Data Science / Project Management	Week 7/8

Consistent with the review cadence established in the Week 4 Risk and Dependency Register, this consolidated report should be reviewed and updated at the start of each subsequent week of the Experience Lab, with High-priority items escalated immediately rather than held for the next scheduled review (PMI, 2021).

References

- AnalystLab Africa. (2026). *HealthConnect Clinic: Risk and dependency register*. Week 4 submission.
- AnalystLab Africa. (2026). *HealthConnect Clinic: Data Science baseline modelling report*. Week 5 submission.
- AnalystLab Africa. (2026). *HealthConnect Clinic: Project execution & progress package*. Week 5 submission.
- AnalystLab Africa. (2026). *HealthConnect Clinic: Model improvement & validation report*. Week 6 submission.
- Dantas, L. F., Fleck, J. L., Cyrino Oliveira, F. L., & Hamacher, S. (2018). No-shows in appointment scheduling – a systematic literature review. *Health Policy*, 122(4), 412–421.
<https://doi.org/10.1016/j.healthpol.2018.02.002>
- Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4), Article 15. <https://doi.org/10.1145/2382577.2382579>
- Project Management Institute. (2021). *A guide to the project management body of knowledge (PMBOK guide)* (7th ed.). Project Management Institute.
- Saeb, S., Lonini, L., Jayaraman, A., Mohr, D. C., & Kording, K. P. (2017). The need to approximate the use-case in clinical machine learning. *GigaScience*, 6(5), gix019.
<https://doi.org/10.1093/gigascience/gix019>
- Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., Chaudhary, V., Young, M., Crespo, J.-F., & Dennison, D. (2015). Hidden technical debt in machine learning systems. In *Advances in Neural Information Processing Systems 28 (NeurIPS 2015)* (pp. 2503–2511).
<https://papers.nips.cc/paper/5656-hidden-technical-debt-in-machine-learning-systems>
- Tabassi, E. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST AI 100-1). National Institute of Standards and Technology. <https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-ai-rmf-10>